

Given: Due:	Further Maths (Statistics) HW 17	Name: MARK SCHEME
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Chapter 2: Poisson, Binomial, Variance & Approximations

A student is investigating the numbers of cherries in a Rays fruit cake. A random sample of Rays fruit cakes is taken and the results are shown in the table below.

Number of cherries	0	1	2	3	4	5	≥ 6
Frequency	24	37	21	12	4	2	0

(a) Calculate the mean and the variance of these data.

(3)

(b) Explain why the results in part (a) suggest that a Poisson distribution may be a suitable model for the number of cherries in a Rays fruit cake.

(1)

The number of cherries in a Rays fruit cake follows a Poisson distribution with mean 1.5

A Rays fruit cake is to be selected at random.

Find the probability that it contains

- (c) (i) exactly 2 cherries,
(ii) at least 1 cherry.

(4)

Rays fruit cakes are sold in packets of 5

(d) Show that the probability that there are more than 10 cherries, in total, in a randomly selected packet of Rays fruit cakes, is 0.1378 correct to 4 decimal places.

(3)

Twelve packets of Rays fruit cakes are selected at random.

(e) Find the probability that exactly 3 packets contain more than 10 cherries.

(3)

Part(a) of this question was poorly answered. A significant proportion of candidates were unable to find the mean and variance of a frequency distribution in part (a). Almost all candidates obtained the sums of fx and fx^2 . However, some candidates did not perform any division. Other candidates divided by seven.

In part (b). There were some candidates who were aware why the evidence in this case (mean $\approx$ variance) suggests that a Poisson distribution may be appropriate. However, many candidates chose instead to mention the conditions under which one might consider a Poisson distribution in the first place, without any knowledge of any actual data.

Part (c), however, was a good source of marks for a very large number of candidates. Even those who had lost the first four marks of the question were able to respond to the challenge of S2 material by recalling the knowledge and performing the skills required by the question.

It was pleasing that almost all candidates provided full detail of the method and working for part (d): this approach is always advisable, but essential when the final answer is given in the question.

Part (e) was also a good source of marks for the vast majority of candidates. A minority of candidates used a Binomial distribution with $n = 10$.

Question Number	EXAM Scheme PRACTICE		Marks
Note : if a correct answer is given with no incorrect working award full marks unless the markscheme says otherwise.			
(a)	Mean = 1.41	B1: Cao Allow 141/100	B1
	Variance = $\frac{343}{100} - 1.41^2$	M1: using $\frac{\sum fx^2}{100} - (\text{their mean})^2$ or $\frac{100}{99} \left(\frac{\sum fx^2}{100} - (\text{their mean})^2 \right)$ oe	M1
		NB Allow the square root of this for the M mark.	
		If no working shown for $\sum fx^2$ then you must see 343, 3.43 or a correct answer	
	= 1.4419 ($s^2 = 1.456$)	A1: awrt 1.44 or 1.46 for s^2	A1
			(3)
(b)	The mean is close to the variance	B1: Cao - allow alternative wording Allow mean equals variance	B1 (1)
(c) (i)	$X \sim \text{Po}(1.5)$ $P(X = 2) = \frac{e^{-1.5} 1.5^2}{2!}$	M1: writing or using $\frac{e^{-\lambda} \lambda^2}{2!}$ or $P(X \leq 2) - P(X \leq 1)$	M1
	= 0.2510	A1: awrt 0.251	A1
(ii)	$P(X \geq 1) = 1 - P(X = 0)$ = $1 - e^{-1.5}$ Or $1 - 0.2231$	M1: writing or using $1 - P(X = 0)$ oe	M1
	= 0.77686....	A1: awrt 0.777	A1
			(4)
(d)	$Y \sim \text{Po}(7.5)$ $P(Y \geq 11) = 1 - P(Y \leq 10)$ = $1 - 0.8622$ = 0.1378 *	B1: Writing Po(7.5) M1: writing $P(Y \geq 11)$ or $1 - P(Y \leq 10)$ oe	B1 M1
		A1: Seeing $1 - 0.8622$ leading to 0.1378 cso (both B1 and M1 awarded)	A1cso (3)
(e)	$A \sim B(12, 0.1378)$	M1: using $(p)^n(1-p)^{12-n}$ where $p = 0.1378$ or 0.138 condone missing nCr	M1
	$P(A = 3) = \binom{12}{3} (0.1378)^3 (0.8622)^9$	M1: $\binom{12}{3} (p)^3 (1-p)^9$, with $0 < p < 1$ Allow 220 or $12C3$ instead of $\binom{12}{3}$	M1
	= 0.1516	A1: awrt 0.152	A1
			(3)
			Total 14

Q2.

The probability of an electrical component being defective is 0.075
The component is supplied in boxes of 120

(a) Using a suitable approximation, estimate the probability that there are more than 3 defective components in a box.

(5)

A retailer buys 2 boxes of components.

(b) Estimate the probability that there are at least 4 defective components in each box.

(2)

In part (a) a significant minority of candidates used a Normal approximation to the Binomial, despite the fact that the Binomial parameters, 120 and 0.075, can only be described as large and small respectively. Those candidates who made their method clear were able to score some marks but those who simply wrote their answers down gained few marks if any. However, a large number of scripts were awarded full marks.

There were some interesting features to the overall response to part (b). Some candidates used an elaborate method that consisted of defining a new Binomial distribution, $B(2, 0.9788)$, and then using the formula for Binomial probability to complete their solution. This was almost always successful, but contrasts (in the amount of work involved) with those candidates who used the multiplication rule for independent events and therefore simply wrote 0.9788×0.9788 or 0.9788^2 .

The most curious aspect of part (b) was the frequency with which candidates retraced their steps by repeating their work for part (a), before proceeding with part (b). Intriguingly, it was not uncommon for candidates to have been incorrect in part (a), but to go on and write a different solution to part (a), correct

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	1st A1 writing or use of Po(9) 2nd M1 for writing or using $1 - P(X \leq 3)$ or this may be implied by an awrt 0.972 using normal approximation. (b) M1 ((their (a))² or 0.979² or 0.9788² or 0.98²	
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A disease occurs in 3% of a population.

(a) State any assumptions that are required to model the number of people with the disease in a random sample of size n as a binomial distribution.

(b) Using this model, find the probability of exactly 2 people having the disease in a random sample of 10 people.

(c) Find the mean and variance of the number of people with the disease in a random sample of 100 people. (2)

A doctor tests a random sample of 100 patients for the disease. He decides to offer all patients a vaccination to protect them from the disease if more than 5 of the sample have the disease.

(d) Using a suitable approximation, find the probability that the doctor will offer all patients a vaccination. (3)

In parts (b) and (c) fully correct solutions were seen in a large majority of candidates. However, there were a small number who failed to spot the change in the value of the parameter n from (b) to (c).

In part (d) most candidates provided a clear and accurate solution. However, there were a few candidates who used a Normal approximation, which is clearly inappropriate in this situation since " n is large and p is small" applies in this case and therefore indicates the use of a Poisson approximation. A small but significant number of candidates used a Binomial distribution despite the question requesting that a suitable approximation be used. A common error from those using a Poisson approximation was the use of $P(X \leq 4)$ instead of $P(X \leq 5)$.

Question Number	Scheme	Marks
(a)	Occurrences of the disease are independent The probability of catching the disease remains constant.	B1 B1 (2)
(b)	$X \sim \text{Bin}(10, 0.03)$ $P(X = 2) = \frac{10 \times 9}{2} (0.03)^2 (0.97)^8 = 0.0317$	B1 M1A1 (3)
(c)	$E(X) = 100 \times 0.03 = 3$ $\text{Var}(X) = 100 \times 0.03 \times 0.97 = 2.91$	B1cao B1cao (2)
(d)	$\lambda = 100 \times 0.03 = 3$ $Y \sim \text{Po}(3)$ $P(Y > 5) = 1 - P(Y \leq 5)$ $= 1 - 0.9161$ $= 0.0839$	B1 (use of) dM1 A1 (3) [10]
Notes		
(a)	B1 independent B1 <u>probability</u> remains <u>constant</u> . One of these must have the context of disease. No context only one correct B0B0 If only one mark awarded give the first B1 SC if they are both correct without context award B1B0	
(b)	B1 for writing or using $B(10, 0.03)$ M1 for writing or using $(p)^2 (1-p)^8 \frac{10!}{2!8!}$ allow $^{10}C_2 \cdot \binom{10}{2}$ etc Allow $P(X \leq 2) - P(X \leq 1)$ A1 awrt 0.0317	
(d)	B1 for <u>using</u> Poisson. Any mean. Common values which imply Poisson used are 0.9665 and 0.8153 dM1 for writing or using $1 - P(X \leq 5)$ - use of binomial gets M0. This is dependent on them being awarded the previous B mark. A1 awrt 0.0839 SC: Use of Normal in (d) Can get B0 M1 A0.- for M1 we must see $1 - P(X \leq 5)$ or $1 - P(X < 5.5)$ or get awrt 0.071	

As part of a selection procedure for a company, applicants have to answer all 20 questions of a multiple choice test. If an applicant chooses answers at random the probability of choosing a correct answer is 0.2 and the number of correct answers is represented by the random variable X .

(a) Suggest a suitable distribution for X .

(2)

Each applicant gains 4 points for each correct answer but loses 1 point for each incorrect answer. The random variable S represents the final score, in points, for an applicant who chooses answers to this test at random.

(b) Show that $S = 5X - 20$

(2)

(c) Find $E(S)$ and $\text{Var}(S)$.

(4)

An applicant who achieves a score of at least 20 points is invited to take part in the final stage of the selection process.

(d) Find $P(S \geq 20)$

(4)

Cameron is taking the final stage of the selection process which is a multiple choice test consisting of 100 questions. He has been preparing for this test and believes that his chance of answering each question correctly is 0.4

(e) Using a suitable approximation, estimate the probability that Cameron answers more than half of the questions correctly.

(5)

This question allowed candidates to score some marks and a variety of solutions were seen. Only the best candidates were able to score full marks and part (b) caused the major problem.

Part (a) was done well by the majority of candidates and errors included either leaving out the 20 and 0.2 or stating a Poisson instead of binomial.

Part (b) was either done well or poorly. A few candidates left this part out whilst others showed that $S = 5X - 20$ for a particular value and not for the general case. In part (c) there was some confusion between X and S . Some candidates could only state $E(X) = 4$ and $\text{Var}(X) = 3.2$. Others assumed S to be uniform. Those candidates who were able to answer the question often scored full marks but the common error for these candidates was to incorrectly calculate $\text{Var}(S)$ (often giving an answer of -0.4).

In part (d) many candidates realised that they had to solve an inequality for X and usually scored full marks, with only a minority making the error of using $P(X \geq 8)$ as $1 - P(X \leq 8)$. Candidates that failed to realise that they needed to solve the inequality in X assumed here that S was normal or at least that they could approximate to the normal.

Part (e) was well done by large numbers of candidates and even those who obtained few marks elsewhere in the question often gave a completely correct solution here. It appeared that candidates have been well trained in answering normal approximation questions. Some answers gave the wrong variance, 40, assuming that the mean and the variance were the same, as with the Poisson. There were a surprising number of candidates who did not show the details of their standardisation which would have been advisable as marks could not be awarded if they had an incorrect value or the variance or mean.



Question Number	Scheme	EXAM PAPERS PRACTICE	Marks
(a)	$X \sim B(20, 0.2)$	M1 A1	(2)
(b)	$S = 4X - 1(20 - X)$ $S = 5X - 20$	M1 A1 also	(2)
(c)	$E(X) = 4, \text{Var}(X) = 3.2$ $E(S) = 5 \times 4 - 20 = 0, \text{Var}(S) = 5^2 \text{Var}(X) = 80$	B1, B1 M1 A1	(4)
(d)	$S \geq 20$ implies $5X - 20 \geq 20$ [So $5X \geq 40$] $X \geq 8$ $P(S \geq 20) = P(X \geq 8) = 1 - P(X \leq 7)$ $= 1 - 0.9679 = \underline{0.0321}$	M1 A1 M1 A1	(4)
(e)	[Let C = no. Cameron gets correct. $C \sim B(100, 0.4)$] $Y \sim N(40, \sqrt{24})$ $P(C > 50) \approx P(Y > 50.5)$ $= P\left(Z > \frac{50.5 - 40}{\sqrt{24}}\right)$ $= P(Z > 2.14...) = 1 - 0.9838 = 0.0162$ or 0.016044.. (awrt <u>0.016</u>) N.B. exact Bin (0.01676...) Poisson approx (0.0526...)	M1A1 M1 M1 A1	(5) [17]
Notes			
(a)	M1 for "binomial" or B(...) A1 for $n = 20$ and $p = 0.2$		
(b)	NB this is a 'show that' so working must be shown M1 for attempt at any correct expression for S that uses 4 and -1 (1 may not be seen) A1 also for correct expression derived. No incorrect working seen and M1 scored.		
(c)	1 st B1 for $E(X) = 4$ seen. Condone $E(S) = 4$. May be implied by correct $E(S)$ or be seen in the calculation for $E(S)$ 2 nd B1 for $\text{Var}(X) = 3.2$ seen. Condone $\text{Var}(S) = 3.2$. May be implied by correct $\text{Var}(S)$ or be seen in the calculation for $\text{Var}(S)$ M1 for a correct formula for $E(S)$ or $\text{Var}(S)$ – follow through their $E(X)$ and $\text{Var}(X)$ may be implied by either answer being correct A1 for 0 and 80 correctly assigned.		
(d)	1 st M1 for an attempt to solve the inequality for X 2 nd M1 for $1 - P(X \leq 7)$		
(e)	1 st M1 for use of normal approx. and mean = 40 1 st A1 for $\text{Var} = 24$ or st. dev = $\sqrt{24}$ May be implied by later work 2 nd M1 49.5 or 50.5 3 rd M1 Standardising using their mean and their sd. If they have not written down a mean and sd then these need to be correct here to award the mark. They must also use 50.5, 49.5 or 50 and find the correct area ie using $1 - P(Z \leq \text{their } 2.14)$, 2 nd A1 for awrt 0.016		